

# Using Extended InGaAs Photodiodes for Detecting Gravitational Waves

J. Briggs, K.A. Strain, S. Hild, A. Bell, B. Barr, and A. Spencer

June 7, 2019

## Abstract

Extended InGaAs is the name given to InGaAs which has been doped with extra indium (more than 53% In) in order to reduce the size of the bandgap, thus making the cut-off wavelength longer. This material is a candidate for the readout photodiodes if the wavelength that gravitational wave detectors are operated at changes to  $2\mu\text{m}$ . Currently, as far as I'm aware, it is impossible to buy extended InGaAs photodiodes that meet the requirements of a detector such as aLIGO. The basic reason why these photodiodes do not meet the performance needed is the lattice structure of the InGaAs, and this leads to questions in the practicalities of creating good enough extended InGaAs photodiodes.

## 1 Introduction

Future upgrades to gravitational wave detectors include increasing the wavelength of the laser used, as the thermal noise of the optical coatings is a limiting source of noise [1] and the creation of coatings which improve as they are cooled is more achievable for  $2\mu\text{m}$  than for 1064 nm or 1550 nm. Changing the lasers wavelength to  $2\mu\text{m}$  means that the current InGaAs photodiodes used also need to be changed. Firstly, these new photodiodes should be able to handle (about) 10 mA of photocurrent. Current gravitational wave detectors operate in the 1-10 mW range at the output (e.g. see reference [2]) and this is due to problems such as having enough light so that the junk light from other modes is a small portion of the total signal. Secondly, they should have a quantum efficiency of more than 99%. The reason for having such a high quantum efficiency requirement is that another future upgrade to gravitational wave detectors is the use of squeezed light to reduce the quantum noise, and losses degrade the amount of squeezing achieved. Finally, the photodiodes should have low  $1/f$  noise. Ground based gravitational wave detectors will be used down to about 10 Hz [3], so the  $1/f$  noise must be below the shot noise of the photocurrent above this frequency. The  $1/f$  noise could potentially be reduced with chopping, however a modulation-demodulation scheme that is compatible with squeezing (as the shot noise would increase by not doing this optimally [4]) would need to be developed and this in itself poses many technical challenges. Photodiode vendors responses vary from saying this diode is impossible to construct to saying that money is the main obstacle. One candidate material is extended InGaAs. Several extended InGaAs photodiodes were tested to see if there was a combination of bias voltage and temperature that would improve the quantum efficiency whilst keeping the noise low. It was found that noise at temperatures between  $0^\circ\text{C}$  and room temperature was almost entirely dependant on reverse bias voltage and did not improve as the device was cooled. It was found that to get adequate noise performance, the diodes needed to be used at a reverse bias voltage that meant their quantum efficiency was far below the requirement. For instance, the FD10D from Thorlabs was measured to have a quantum efficiency at 1550 nm when used at its full reverse bias, and at this bias the  $1/f$  noise only became less than a typical shot noise above 1 kHz

## 2 PIN InGaAs Photodiodes

PIN photodiodes use a positive-intrinsic-negative semiconductor structure to create a device that when light of sufficient energy, energy greater than the band gap of the semiconductor, is incident on it an electron-hole pair is produced. Pairs that are produced in the intrinsic region result in electrons drifting towards the N-type semiconductor and holes drifting towards the P-type. If the electron-hole pair is created in the N region, the electrons stay in the conduction band meanwhile the holes diffuse to the P

layer (and vice versa for electron hole pairs created in the the P layer). By applying a reverse bias, the depletion region is widened resulting in the conversion of light to current being faster. The ratio of total photons incident on the photodiode to photons which produce electrons-hole pairs is called quantum efficiency.

Photodiodes produce a dark current  $I_d$  which depends on the reverse bias. The dominant source of dark current depends on the regime of reverse bias that the diode is under and can be separated into three sources: diffusion current  $I_{diff}$ ; generation-recombination current  $I_{GR}$  and quantum tunnelling current  $I_{tun}$  [5],

$$I_d = I_{tun} + I_{diff} + I_{GR}. \quad (1)$$

$I_{tun}$  arises due to electrons being able to tunnel across the band gap, similar to the breakdown voltage of a diode.  $I_{diff}$  arises due to the diffusion of minority carriers into the intrinsic region.  $I_{GR}$  is due to the generation and recombination of electron hole pairs in the intrinsic region, and the random nature of the generation and recombination of electron hole pairs is the source of  $1/f$  noise[6]. This noise is sometimes referred to as random telegraph or popcorn noise. Electrons and holes enter what are known as trap states, states which ideally would not exist, and the amount of time spent there before leaving and recombining with electrons and holes is random. These states arise due to lattice mismatching, which is why its hard to make low noise extended InGaAs photodiodes. The spectrum of one trap causing generation-recombination of electrons and holes is a Lorentzian which when many traps are near each other can look like a  $1/f$  spectrum[7][8]. Ignoring DC offsets (which appears as a delta function in the power spectrum), the power spectrum for one trap is

$$S(\omega) = \frac{1}{\pi} \frac{\sigma\tau}{(\sigma + \tau)^2} \frac{\frac{1}{\sigma} + \frac{1}{\tau}}{\omega^2 + \left(\frac{1}{\sigma} + \frac{1}{\tau}\right)^2} \quad (2)$$

where  $\sigma$  and  $\tau$  are time constants for an electron moving between the trapped state and the free state. This description of  $1/f$  noise is known as McWorters model, and the result is illustrated in figure 1.

InGaAs is a widely used semiconductor for fabricating photodiodes. Typically they are sensitive up to wavelengths around 1550nm, however longer wavelength sensitivity can be achieved by increasing the amount of indium doping. These photodiodes are often called ‘extended’ and can be sensitive for wavelengths up to 2.6 $\mu$ m however the increased level of indium doping causes more mismatches in the lattice, causing more trap states and leading to a greater amount of  $1/f$  noise. This  $1/f$  noise can be mitigated by modulating the bias voltage [7], however modulation of the revers bias voltage can lead to an increase the shot noise [4].

### 3 Technical Problems In Creating Extended InGaAs Photodiodes.

One of the main issues faced when creating an extended InGaAs photodiode is the lattice mismatch between the InGaAs and the InP substrate that its grown on (GaAs can be used as a substrate, while being cheaper and more mechanically robust it increases the amount of mismatching). Above 53% indium, a buffer layer of InGaAs with less indium in between the absorbing layer and the substrate can be used to mitigate the lattice mismatch slightly, but the dislocations still give rise to trap states which cause the generation and recombination of charge carriers [9]. This is the source of the large amounts of dark current found in extended InGaAs, as well as the  $1/f$  noise.

Creating a photodiode which has an internal quantum efficiency of nearly 100% is hard to achieve due to the lattice mismatch making it hard to make the photodiode thick. The parameters that affect the internal quantum efficiency are the absorption coefficient of the extended InGaAs and the amount of the extended InGaAs the light must travel through. For  $\text{In}_{0.8}\text{Ga}_{0.2}\text{As}$ , the absorption coefficient,  $\alpha$ , at 2.5 $\mu$ m light is approximately  $10^4 \text{ cm}^{-1}$  [10]. For 10mW ( $I_0$ ) of light and 99% quantum efficiency (giving  $I_x = 0.01 \times I_0$ ), we can find the thickness  $x$  of the InGaAs using the Beer-Lambert law,  $\frac{I_x}{I_0} = e^{-\alpha x}$  and find that  $x = 4.6\mu\text{m}$ . Although this is just a rough calculation, it does give one an idea of how thick the material needs to be. This assumes all the photons absorbed make it out of the junction as

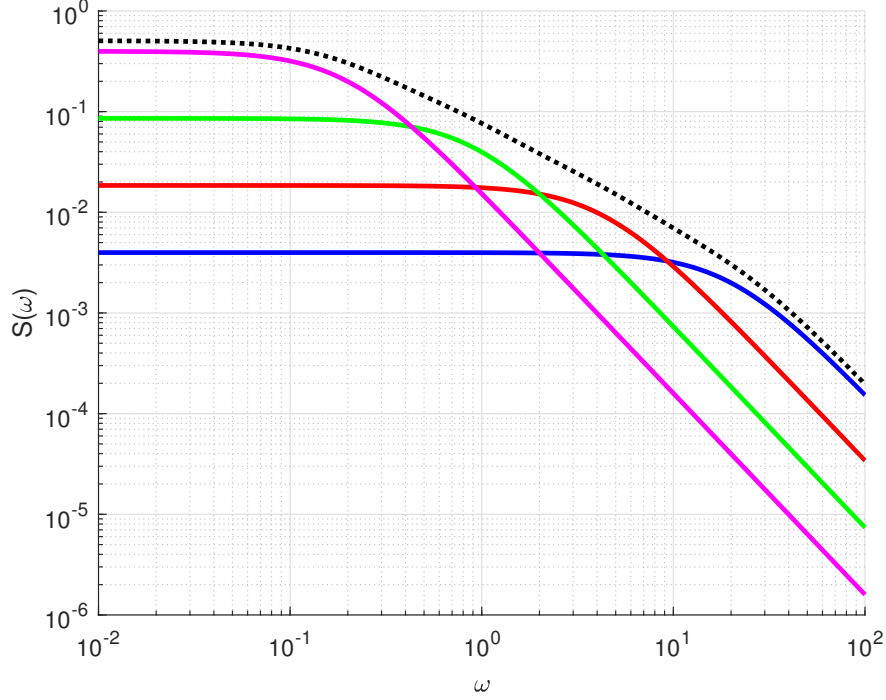


Figure 1: By combining several Lorentzian spectra,  $1/f$  noise is produced. The dashed line shows the sum of the coloured lines, which has a region where the spectrum is  $1/f$ . The coloured lines are determined by equation 2, with  $\sigma = \tau$  and computed for  $\sigma \in \{0.1, 0.46, 2.15, 10\}$ .

electrons/holes, which may not be the case as without a high enough bias the electrons will fail make it out of the junction before recombining with a hole.

The layer of InGaAs could only be about 100 nm thick for there to be a low amount of strain between the InP and the InGaAs layer. Above this thickness, dislocations due to the InGaAs not being lattice matched to the InP are a problem. The thicker the InGaAs layer is, the more strain relief is required between the InP substrate and the InGaAs layer. The more strain relief there is, the more dislocations there will be which will lead to more  $1/f$  noise and dark current. There is not a way around this. The critical thickness is the thickness above which the semiconductors quality degrades significantly, and the higher the ratio of indium to gallium, the thinner this critical thickness is [11]. Putting 100 nm into equation 3 gives an internal quantum efficiency of about 10%. Another design challenge is InGaAs is not a good conductor of heat so a thicker amount of InGaAs can lead to lower power handling capabilities as the photodiode will saturate at a lower photocurrent [12]. Once the internal quantum efficiency problem has been solved, an anti reflection coating would need to be developed. For  $\text{In}_{0.8}\text{Ga}_{0.2}\text{As}$ , the refractive index  $n$  for  $2.5\mu\text{m}$  light is approximately 3.6 [10]. The film would need a refractive index that was  $n_f = \sqrt{3.6}$  and be 600 nm thick. The idea of using this layer as a passivation layer is presented in [13].

## 4 Apparatus

To measure the dark current that different photodiodes produce under different reverse bias voltages and easily obtainable temperatures, a dark box containing a peltier module was constructed. The photodiode was held in place against a piece of aluminium that was also in contact with the peltier. An LM35 temperature sensor was used to measure the temperature of the aluminium. The dark current of the photodiodes was expected to be of the order nanoamps so a  $1\text{ M}\Omega$  resistor was used in the transimpedance amplifier. Due to this choice of resistor a TL071 was used, as shown in figure 2. The bias voltage was provided by a lm7815 voltage regulator and since not much current was being drawn a variable resistor was used to set the bias voltage. A  $1000\mu\text{F}$  electrolytic capacitor was used to reduce the bias voltage noise.

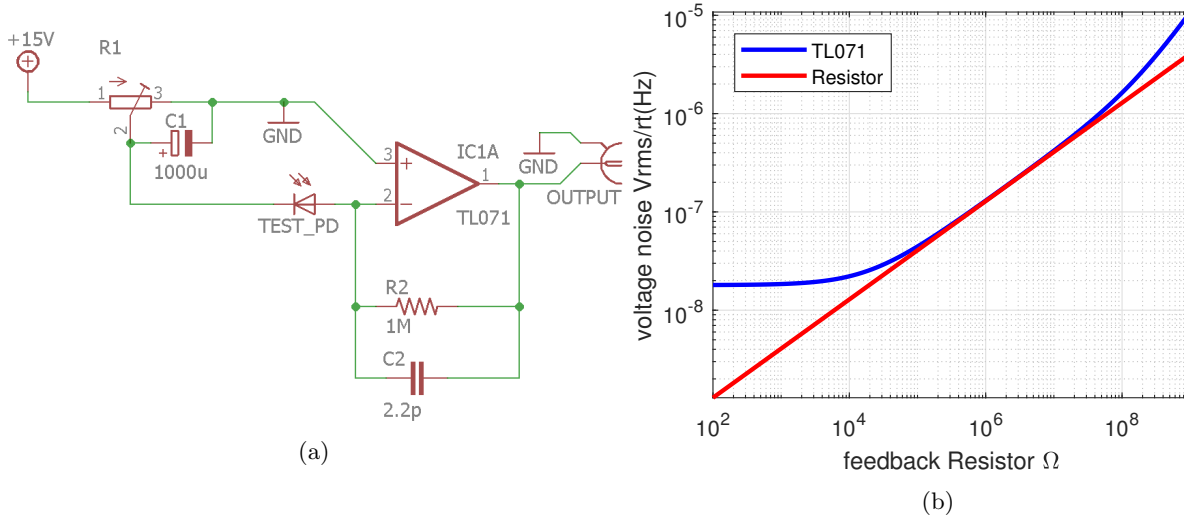


Figure 2: The transimpedance amplifier used to measure the dark noise of a variety of photodiodes is shown in figure 2a. The voltage noise at the output of the transimpedance amplifier compared to the Johnson noise of a resistor is shown in 2b. The TL071 is optimised for between  $100\text{ k}\Omega$  to  $10\text{ M}\Omega$ .

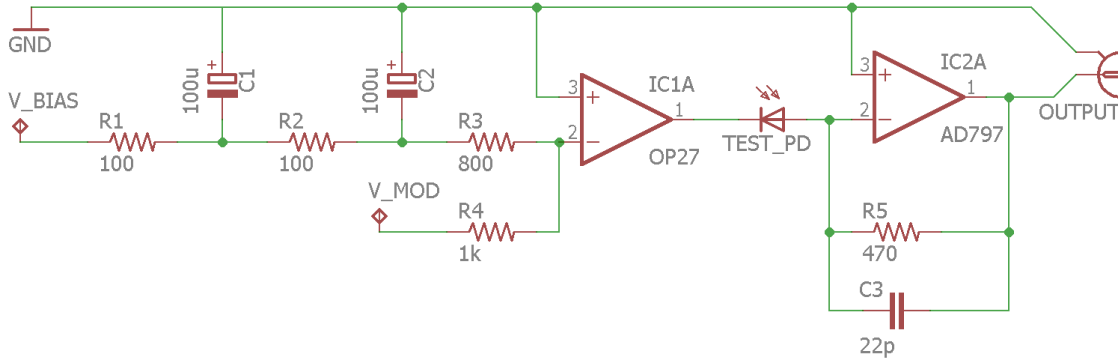


Figure 3: Transimpedance amplifier used to measure the quantum efficiency and linearity of photodiodes for up to  $20\text{ mW}$  of  $1550\text{ nm}$  laser light.

To measure the relative quantum efficiency and linearity of photodiodes at light powers up to  $20\text{ mW}$ , a  $1500\text{ nm}$  wavelength laser (PPCL300) from pure photonics was used. The transimpedance amplifier shown in figure 3 was built to test the photodiodes under different bias voltages. A low passed bench top power supply was used to provide the DC bias voltage. Without filtering it was found that the noise of the power supply was the limiting source of noise when there was laser light on the extended InGaAs photodiodes, as the impedance of the photodiodes decreased when light was on the photodiode. To prove that the increase in photocurrent due to increasing the the reverse bias, the shot noise of the photocurrent was measured. To do this an amplitude stabilised laser was required. The photodiode used for this was a C30641gh and it was assumed this diode behaved well. The apparatus is shown in figure 4. To make the measurements, an Stanford Research Systems SRS785 was used.

## 5 Experimental Results from Testing Extended InGaAs Photodiodes.

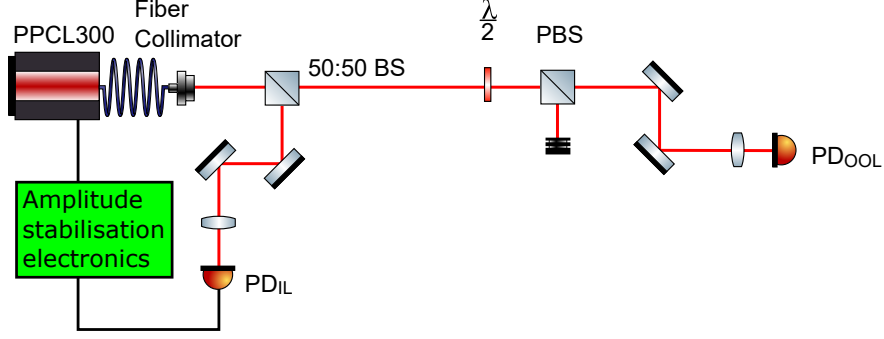


Figure 4: Layout of experiment.  $PD_{IL}$  is the in loop photodiode (C30641gh) and  $PD_{OOL}$  is the photodiode being tested.

The FD10D is a 1 mm photodiode sold by Thorlabs, with a cut-off wavelength of  $2.6\text{ }\mu\text{m}$ . Its dark noise is shown in figure 5, showing that for use in a gravitational wave detector, running this diode with a reverse bias much higher than  $0.6\text{ V}$  means that  $1/f$  noise will be a significant problem. Figure 7 shows why one would want to increase the bias as much as possible, the amount of photocurrent generated increases by 17% for this particular diode when the reverse bias is turned up to its maximum datasheet value. The absolute quantum efficiency was measured for this diode, and taking into account reflections from the glass window in front, to be 82% at  $1.8\text{ V}$  reverse bias. At  $0.6\text{ V}$  reverse bias the  $1/f$  noise is below the target, and the quantum efficiency was measured to be 70%. The maximum photocurrent this diode produced at  $0.6\text{ V}$  reverse bias before starting to saturating was  $15\text{ mA}$ .

To ensure that the bias voltage for the switchable photodiode design met the noise requirements, a fixed bias voltage with noise at  $10\text{ Hz}$  to be  $50\text{ nV}_{rms}/\sqrt{\text{Hz}}$  was used to cross check the measurement of dark noise for the FD10D. This requirement was set by assuming that the dynamic impedance of the test photodiodes would be of the order  $100\text{ k}\Omega$ , and the smallest amount of current noise that is measurable at the inverting input of the TL071 in this circuit is  $0.5\text{ pA}_{rms}/\sqrt{\text{Hz}}$ , so multiplying these two quantities gives the bias noise requirement. By having a socket rather than a solder joint and a standard voltage regulator to provide the bias, there is potential to have of bias voltage noise of several  $100\text{ uV}_{rms}/\sqrt{\text{Hz}}$ . An AD587 with 2 low pass filters was used to create the quiet bias voltage, this is shown in figure 8. There was no difference in noise between using the switchable photodiode test circuit and the quiet bias voltage circuit, as shown in figure 9, so we can rule out bias voltage noise being the cause of the  $1/f$  noise. If the bias voltage produced  $1/f$  noise and the photodiode also produced  $1/f$  noise, then you would see a corner in the spectrum, as well as a generally messy looking spectrum.

This behaviour was not unique to the FD10D - several from Laser Components were also tested. The main problem was the  $1/f$  noise becoming large, forcing one to use a small reverse bias. A small reverse bias results in electrons-hole pairs failing to make it out of the junction before recombining, leading to a smaller quantum efficiency and reduced power handling capabilities. Although the FD10D did manage to handle a reasonable amount of power, the quantum efficiency was still far from the target. The Laser Components extended InGaAs range performed poorly in both noise and power handling terms (see figures 10 and 11).

The forward current of the IG24x500S4i was measured using the apparatus shown in figure 2 and modelled using

$$I = I_s(e^{-qV/nk_B T} - 1) \quad (3)$$

where  $I_s$  is the saturation current,  $q$  is the electron charge,  $V$  is the voltage across the diode,  $n$  is the ideality factor,  $k_B$  is the Boltzmann constant and  $T$  is the temperature. The ideality factor depends on the mechanism by which the generation-recombination current is created by. An ideality factor of 1 means that the process is driven by the existence of trap states. This is shown in figure 12. Fitting for  $I_s$  and  $n$  gives the saturation current to be  $0.36\text{ mA}$  and the ideality factor to be  $1.02$ .

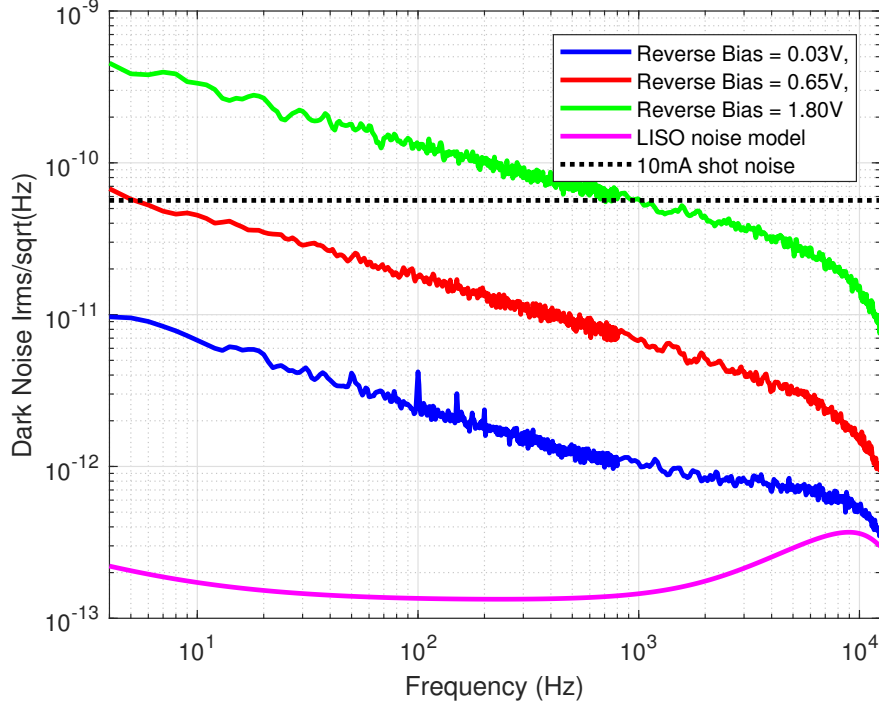


Figure 5: The dark noise of the FD10D at different reverse biases. The maximum datasheet value is 1.8V, above this voltage electrons can start tunnelling across the band gap. The  $1/f$  noise can be explained by McWorters Model of noise in semiconductors. These measurements were made at room temperature.

## 6 Conclusion

Current extended InGaAs photodiode technology is not advanced enough to be useful in the readout schemes of future gravitational wave detectors given that simple PIN extended InGaAs photodiode diodes at room temperature are either too noisy, cannot handle enough power or do not have a high enough quantum efficiency. To overcome  $1/f$  noise either a low bias is required which reduces the quantum efficiency, or a modulation/demodulation i.e. chopping is required. Both of these techniques result in photons not being detected which detracts the amount of squeezing one can achieve. Suppliers of extended InGaAs photodiodes do not offer much hope when asked if a photodiode that meets LIGO's requirement can be produced. A possible material for detecting  $2\mu\text{m}$  light is germanium however these diodes have low shunt resistances and high dark current. HgCdTe could also be used as reference [?] indicates that detectors with 94% quantum efficiency and have noise equivalent powers even at room temperature well below the shot noise requirement of LIGO can be created, however this needs further investigation.

## 7 Acknowledgements

I would like to thank Andrew Long for our useful discussion about InGaAs and Sebastian Steinlechner for providing us with some Laser Components photodiodes to test.

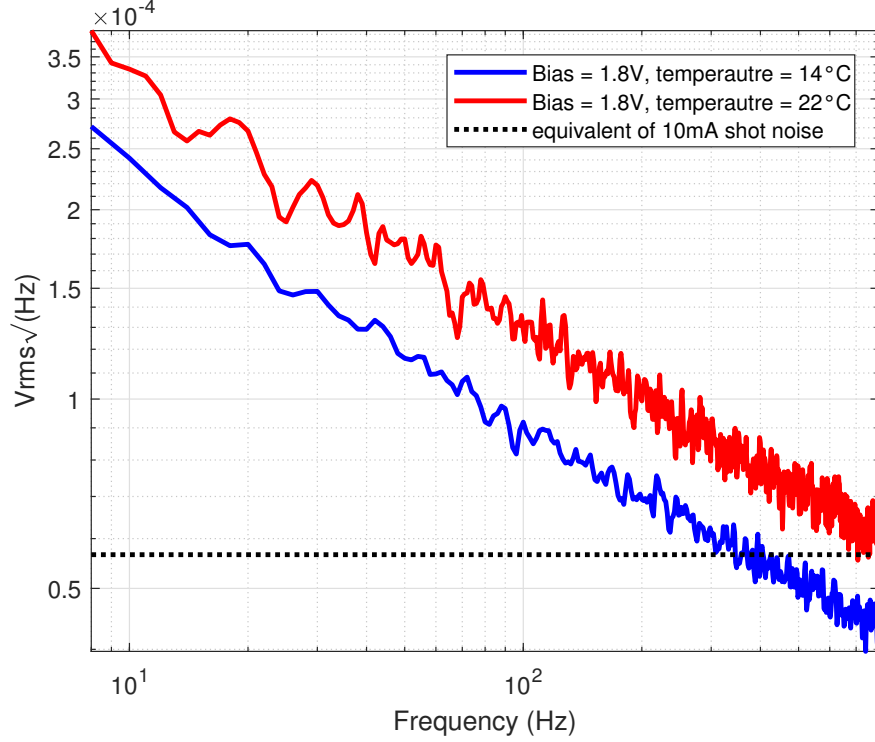


Figure 6: The dark noise of the FD10D with a reverse bias of 1.8V at room temperature and 14°C. Although there is a slight reduction in noise, this comes at the expense of less quantum efficiency near the cut off wavelength.

## References

- [1] B. Lantz, S. L. Danilishin, S. Hild, Gustafson E., D. Coyne, V. Quetschke, G. D. Hammond, R. Adhikari, M. Evans, and R. Bassiri. Instrument science white paper 2018. LIGO DCC.
- [2] M Prijatelj, J Degallaix, H Grote, J Leong, C Affeldt, S Hild, H Lck, J Slutsky, H Wittel, K Strain, and K Danzmann. The output mode cleaner of GEO 600. *Classical and Quantum Gravity*, 29(5):055009, feb 2012.
- [3] Gregory M Harry and the LIGO Scientific Collaboration. Advanced LIGO: the next generation of gravitational wave detectors. *Classical and Quantum Gravity*, 27(8):084006, apr 2010.
- [4] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [5] S. Forrest. Performance of  $\text{In}_x\text{Ga}_{1-x}\text{As}_y\text{P}_{1-y}$  photodiodes with dark current limited by diffusion, generation recombination, and tunneling. *IEEE Journal of Quantum Electronics*, 17(2):217–226, feb 1981.
- [6] Frank Seifert. *Power stabilization of high power lasers for second generation gravitational wave detectors*. PhD thesis, Hannover, Univ., Diss.; 2009, 2010.
- [7] Eric A.M. Klumperink, A.P. van der Wel, J.S. Kolhatkar, E. Hoekstra, Cora Salm, Hans Wallinga, and Bram Nauta. *Reduction of 1/f Noise by Switched Biasing: an Overview*, pages 307–315. Technology Foundation (STW), 11 2005.
- [8] Stefan Machlup. Noise in semiconductors: Spectrum of a two-parameter random signal. *Journal of Applied Physics*, 25(3):341–343, mar 1954.
- [9] Lars Zimmermann, Joachim John, Stefan Degroote, Gustaaf Borghs, Chris Van Hoof, and Stefan Nemeth. Extended wavelength InGaAs on GaAs using InAlAs buffer for back-side-illuminated short-wave infrared detectors. *Applied Physics Letters*, 82(17):2838–2840, apr 2003.

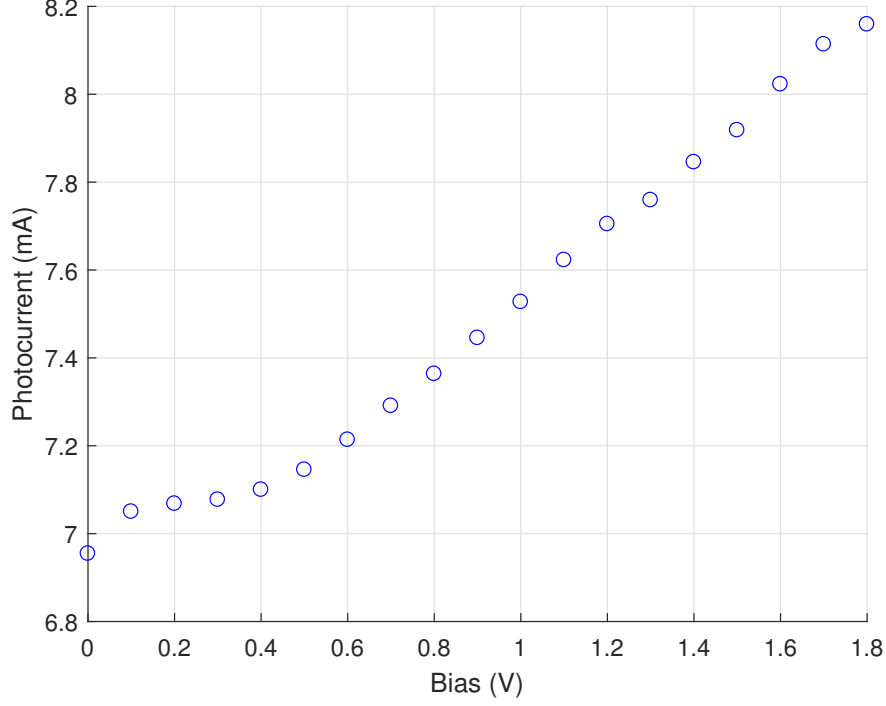


Figure 7: A 1550nm laser was used to illuminate the FD10D at room temperature. As the reverse bias is increased, the photocurrent out of the FD10D increases by about 17%.

- [10] M. S. Alam, M. S. Rahman, M. R. Islam, A. G. Bhuiyan, and M. Yamada. Refractive index, absorption coefficient, and photoelastic constant: Key parameters of InGaAs material relevant to InGaAs-based device performance. In *2007 IEEE 19th International Conference on Indium Phosphide & Related Materials*. IEEE, may 2007.
- [11] Takayoshi Anan, Kenichi Nishi, and Sigeo Sugou. Critical layer thickness on (111)b-oriented InGaAs/GaAs heteroepitaxy. *Applied Physics Letters*, 60(25):3159–3161, jun 1992.
- [12] Xiaowei Li, Ning Li, S. Demiguel, J.C. Campbell, D. Tulchinsky, and K.J. Williams. A comparison of front- and backside-illuminated high-saturation power partially depleted absorber photodetectors. *IEEE Journal of Quantum Electronics*, 40(9):1321–1325, sep 2004.
- [13] M. López, H. Hofer, K. D. Stock, J. C. Bermúdez, A. Schirmacher, F. Schneck, and S. Kck. Spectral reflectance and responsivity of ge- and InGaAs-photodiodes in the near-infrared: measurement and model. *Applied Optics*, 46(29):7337, oct 2007.

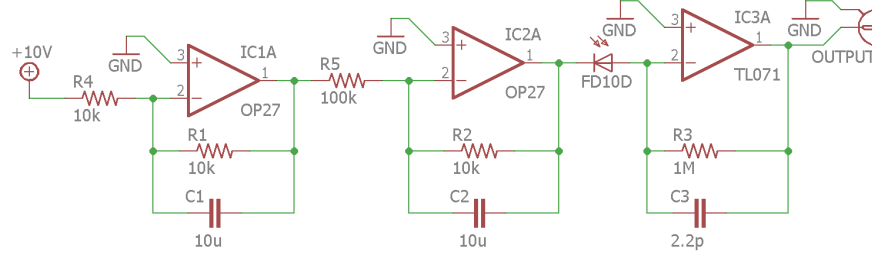


Figure 8: Low noise fixed bias voltage dark noise measurement circuit. The 10V reference is generated by an AD587, in this circuit the noise is limited by the voltage noise of the second OP27, but it still reaches  $5 \text{ nV}_{rms}/\sqrt{\text{Hz}}$  which is 10 times smaller than the requirement.

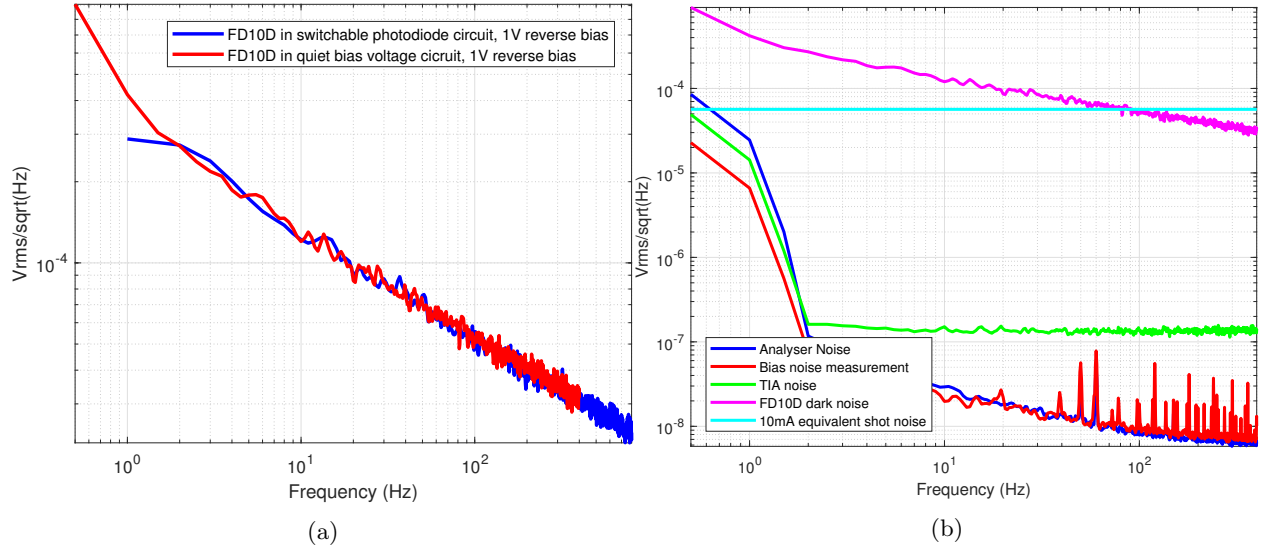


Figure 9: The noise at the output of both dark noise test circuits were cross checked and found to be in agreement as shown by figure 9a. The noise bias voltage in the quiet bias voltage circuit was found to meet the requirements as shown by figure 9b.

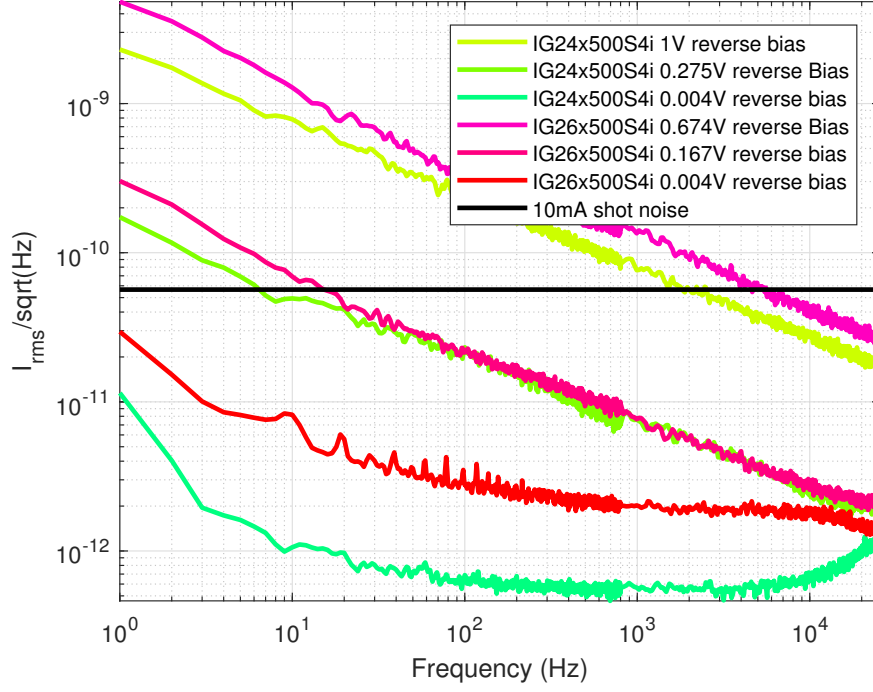


Figure 10: The dark noise of several two different extended InGaAs photodiodes from laser components measured at room temperature.

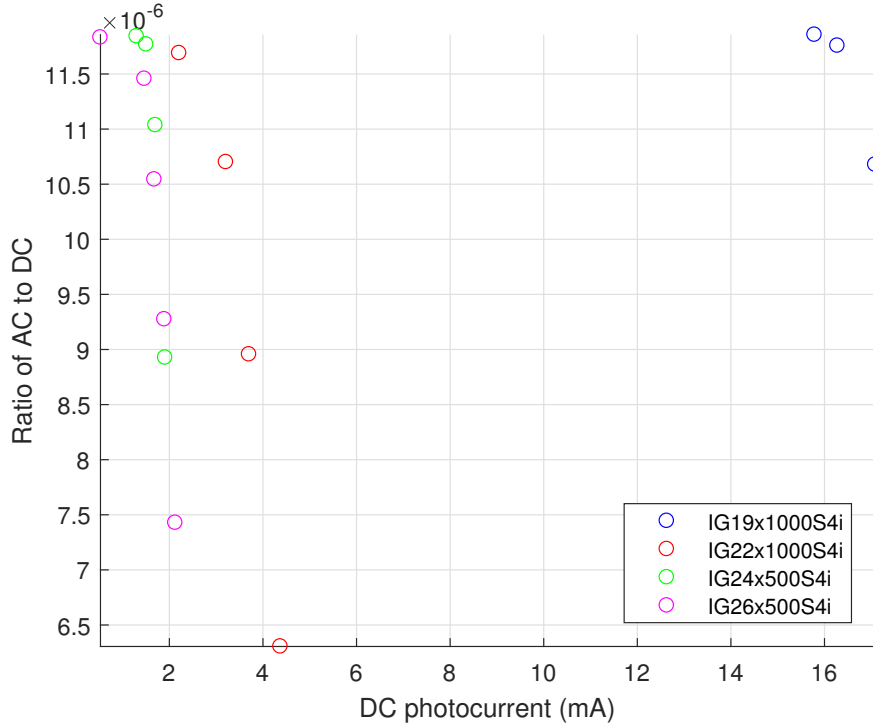


Figure 11: The amount of power that several different Laser Components photodiodes can handle before the signal out of them stops being linear while keeping the  $1/f$  noise below the shot noise of a 10 mA current. A small modulation was applied to an amplitude stabilised laser, the size of this modulation was compared to the DC photocurrent and the ratio of them was plotted. The drop off shows that the DC signal is getting bigger while the AC signal is being saturated.

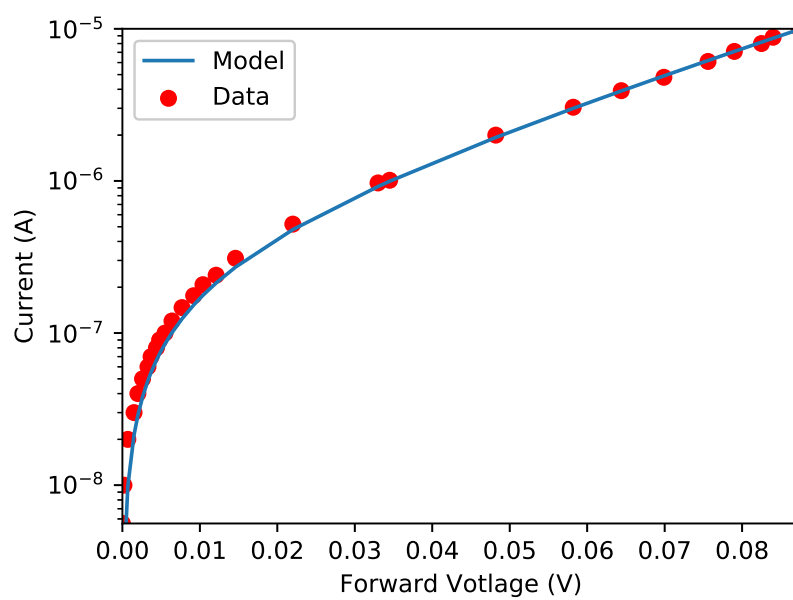


Figure 12: The forward IV curve of the IG24x500S4i.